

Section A: Pure Mathematics (40 marks)

1 Show that $x^2(x+2)^2 - 1 = (x+1)^2(x+1+\sqrt{2})(x+1-\sqrt{2})$. **[1]**

Hence, without using a calculator, solve the inequality $\frac{1}{(x+2)^2} \geq x^2$. **[3]**

Deduce the range of values of x for which $\frac{1}{(e^x+2)^2} \geq e^{2x}$. **[2]**

- 2** A certain economy comprises of three industries that produce three products: X , Y and Z . In a given week, it produces $\$x$ worth of X , $\$y$ worth of Y and $\$z$ worth of Z .
The production of $\$1$ worth of any particular product requires the use of some of its own product, and some of the two other products, as shown in the following table.

requires the use of	The production of $\$1$ worth of		
	X	Y	Z
X	$\$0.10$	$\$0.25$	$\$0.20$
Y	$\$0.30$	$\$0.40$	$\$0.50$
Z	$\$0.10$	$\$0.15$	$\$0.10$

For instance, to produce $\$1$ worth of X , one needs to use $\$0.10$ worth of X , $\$0.30$ worth of Y and $\$0.10$ worth of Z .

Other than providing internally for the three industries, each industry must also meet its external demand. During the given week, the economy has an external demand of $\$50000$ worth of X , $\$75000$ worth of Y , and $\$125000$ worth of Z .

- (i)** By considering how the industry for X meets both its internal provision and external demand, show that

$$0.9x - 0.25y - 0.2z = 50000. \quad \text{[2]}$$

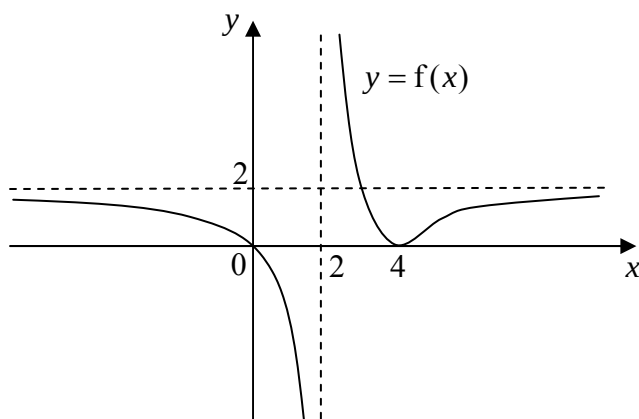
- (ii)** By forming a system of linear equations, find the values of x , y and z , correct to the nearest integer, in order that the entire economy exactly satisfies both the internal provisions and the external demands for that given week. **[4]**

[Turn over]

- 3 The point P in an Argand diagram represents the variable complex number z and the point A in the first quadrant represents the fixed complex number a . Sketch, on separate diagrams, the locus of P in each of the following cases, marking clearly the relationship between the locus and the point A .

- (i) $|z - a| = |a|$, [2]
 (ii) $|z + 2a| = |z|$, [2]
 (iii) $\arg(z - a) = \arg(a) + \frac{1}{2}\pi$. [2]

- 4 The graph of $y = f(x)$ has a minimum turning point at $(4, 0)$ and passes through the origin. The lines $x = 2$ and $y = 2$ are asymptotes to the graph, as shown in the diagram below.



- (i) State the range of values of x for which the graph of $y = \frac{1}{f(x)}$ is decreasing. [1]
 (ii) State the range of values of x for which the graph of $y = f'(x)$ is below the x -axis. [2]
 (iii) Sketch the graph of $y^2 = f(x)$, showing clearly the equations of all asymptotes and the shape of the graph at the origin. [3]
 (iv) Sketch the graph of $y = f(|x|) + 2$, showing clearly the equations of all asymptotes and the coordinates of the stationary points. [3]

- 5 (a) In the expansion of $(1 + 2x + 3x^2)^n$ in ascending powers of x , the coefficient of x^2 is 105. Given that n is a positive integer, find the value of n . [3]

- (b) Obtain the expansion of $\frac{1-4x}{3+x^2}$ up to and including the term in x^4 .
Find the set of values of x for which the expansion is valid. [4]

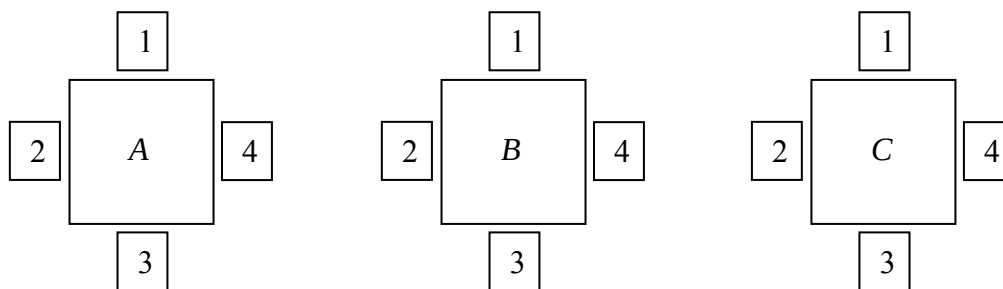
- (c) The function $y = f(x)$ satisfies the differential equation $2y \frac{dy}{dx} = y^2 - 1$ and its graph passes through the point $(0, 3)$.

Show that when $x = 0$, $\frac{d^2y}{dx^2} = \frac{20}{27}$. [3]

Find the Maclaurin's series of y in ascending powers of x , up to and including the term in x^3 . [3]

Section B: Statistics (60 marks)

- 6 A group of 12 students consisting of 6 girls and 6 boys are to be seated at tables A, B and C, each with 4 chairs numbered 1, 2, 3 and 4 as shown below.



Find the number of ways this can be done if

- (i) there is no restriction, [1]
(ii) there are 2 girls and 2 boys at each table, [3]
(iii) there are 2 girls and 2 boys at each table and students of the same gender are seated opposite each other. [2]

[Turn over]

- 7 A company has 6 departments and a total of 240 staff. Each department has a different number of staff. The management, concerned by a large turnover, decides to survey a sample of 30 staff to seek their opinions on the working conditions in the company.

Two methods are suggested on how the sample can be selected.

Method 1: Use Simple Random Sampling

Method 2: Use Stratified Sampling

- (i) Describe how each method can be carried out. [4]
- (ii) Give one reason for choosing
- (a) Method 1 over Method 2, [1]
- (b) Method 2 over Method 1. [1]
- 8 The number of flaws in a roll of a particular design of wallpaper is known to follow a Poisson distribution with mean 0.15.
- (i) Find the probability that, of two randomly chosen rolls of wallpaper, one has exactly 1 flaw and one has more than 1 flaw. [3]
- (ii) Using a suitable approximation, find the probability that in a random sample of 50 rolls of wallpaper, there is an average of more than 0.3 flaw per roll. [3]
- 9 From records, a shot putter has an average throwing distance of 8.92 metres. His coach taught him a new technique and after three months of training, his distance, in metres, on 10 different occasions is as follows.

9.11 8.89 9.02 9.08 9.10 8.90 9.05 8.95 8.87 8.86

The coach is interested to find out if the shot putter has improved with the new technique.

- (i) State, giving your reasons, whether a z-test or t-test should be used. [2]
- (ii) Carry out the test in (i) at the 5% significance level, stating any assumption necessary for validity. [5]

- 10** The probability that a hockey team wins any match is $\frac{1}{2}$ and the probability that it loses any match is $\frac{1}{6}$. Three points are awarded for a win, one point for a draw and no point for a defeat. The team plays four matches and the outcome of each match is independent of the other matches.
- (i) Show that the probability that the team has exactly one draw and exactly one defeat is $\frac{1}{6}$. [1]
- Find the probability that the team
- (ii) wins the first match and goes on to win exactly one other match, [3]
- (iii) wins exactly one match, given that it obtains four points. [4]
- 11** In a certain factory, it is known that 5% of all articles produced are defective.
- (i) Show that the probability of a randomly chosen sample of size 10 containing exactly one defective article is 0.315, correct to 3 significant figures. [3]
- (ii) 80 random samples, each of size 10, are drawn. Using a suitable approximation, find the probability that at least 25 but fewer than 35 samples will each contain exactly one defective article. [5]
- 12** A factory manufactures a large number of containers with press-on lids. The diameter of a randomly chosen lid is X cm, where X follows a normal distribution with mean 12.1 and standard deviation 0.03. The diameter of the top of a randomly chosen container into which the lid is pressed is Y cm, where Y follows an independent normal distribution with mean 12.0 and standard deviation 0.03.
- (a) Find the probability that the sum of the diameters of 3 randomly chosen lids and twice the diameter of the top of a randomly chosen container is less than 60.5 cm. [4]
- (b) A lid and a container are chosen randomly and paired. A pairing is accepted if $0.02 < X - Y < 0.17$ and discarded otherwise.
- (i) Find the probability that a pairing is accepted. [3]
- (ii) Estimate the number of trials required to obtain 2000 accepted pairings. [2]

[Turn over]

- 13** A stationery retailer supplies pens to offices. In order to encourage customers to buy in bulk, the stationery retailer came up with the following price scheme.

Number of pens ordered, x	5	10	20	40	80	160
Unit price in dollar, y	1.50	1.20	1.10	0.99	0.90	0.80

- (i) Calculate the value of the product moment correlation coefficient. **[1]**
- (ii) Give a sketch of the scatter diagram for the data and hence comment on the value of the product moment correlation coefficient found in (i). **[3]**
- (iii) State, with a reason, which model of the form $y = a + bw$ is appropriate, where a and b are constants, $b > 0$ and w is as follows.

A: $w = x^2$,

B: $w = \frac{1}{x}$,

C: $w = \ln x$.

[2]

Using the appropriate model selected above, find

- (iv) the value of the product moment correlation coefficient, **[1]**
- (v) the equation of the regression line of y on w , and hence obtain an estimate of the number of pens ordered when a unit price of \$1.40 is quoted.
Comment on the reliability of your answer. **[3]**